ADVANCED INDUSTRIAL ORGANIZATION

LECTURE 4

Product Differentiation

Product positioning, quality, market structure and empirical demand

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Study notes based on the lectures of Hui Shen

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Reading guide

These notes follow Lecture 4, based on Hui Shen's slides dated October 21, 2026. The section and subsection structure follows the course outline; smaller explanatory headings develop the material within that structure. Progressive builds are consolidated without omitting their additional arguments, examples or calculations.

Original **Model numbers** identify the classroom models; each is followed by a descriptive name. Course figures are redrawn with their axes and economic interpretation. Numbered takeaways separate the conclusions, their mechanisms and the assumptions that support them.

A model is read in the order

$$\text{context} \longrightarrow \text{choices} \longrightarrow \text{optimization} \longrightarrow \text{equilibrium} \longrightarrow \text{meaning}.$$

For a sequential model, start with the last decision and work backwards. For a derivative, identify the variable being changed and what remains fixed. A first-order condition is followed by the relevant feasibility, boundary and deviation checks.

Definitions and results

Blue boxes state the conclusions and the assumptions under which they hold. A displayed optimum is a conclusion about the specified model, rather than a rule for every industry.

Economic mechanism

Green boxes explain the incentives and distinguish a direct effect from an equilibrium response. In particular, a price reduction is a transfer unless it also changes quantities, matching or other real outcomes.

Examples, exercises and lecture cases

Purple boxes contain calculations or applications. Historical cases retain their original period and numerical context. An exercise's hypothetical assumptions are distinguished from historical facts.

Conditions and boundary cases

Amber boxes keep the qualifications that matter for a result: market coverage, nonnegative quantities, integer entry, credible commitment, or an approximation that neglects a firm's effect on an aggregate.

Derivations are beside the model. Boxes labelled "Lecture derivation" develop the classroom calculations and proofs, including those placed in the slides' appendix. Intermediate algebra and explanatory steps are included where needed; no separate mathematical appendix is required to follow the argument.

Instructor stars. The symbol $\star$ preserves an instructor-starred topic. It identifies the course's starred or supplementary material, not an additional importance ranking. Starred sections retain their star in the contents, and the notes state when the slides identify them as skipped.

Notation

Symbol	Meaning and local convention
$x \in [0,1]$, l_i	Consumer ideal point and product location.
τ, t	Transport/mismatch parameters; the escalator exercise uses both.
r, c	Baseline willingness to pay and marginal production cost.
$\widehat{x}$, $\widehat{\theta}$	Consumer indifferent between two products.
$s_i \in [\underline{s}, \overline{s}]$	Product quality. Here s_i is not a market share.
$\theta \in [\underline{\theta}, \overline{\theta}]$	Consumer willingness to pay for an extra quality unit.
$\Delta = s_2 - s_1$	Quality gap, positive when firm 2 has higher quality.
$\overline{v}_i$, ϵ_{ij}	Systematic utility and consumer j's idiosyncratic utility for product i.
$\mu > 0$, γ_E	Logit taste-shock scale and Euler's constant.
α_i, α_0	Inside-product and outside-option market shares.
β, $\gamma > 0$	Observed-characteristic coefficients and price disutility coefficient.
ξ_i, ω_i	Unobserved demand and marginal-cost characteristics.
$\sigma \in [0,1)$	Nested-logit correlation parameter; not a CES elasticity here.
$\alpha_{i\|g}$	Product i's share conditional on choosing nest g.

Keep the conditioning visible. In the pricing stage, locations or qualities are fixed. In the preceding product-choice stage, firms anticipate how equilibrium prices respond. A conditional choice probability for one consumer type must be averaged over types before it becomes an aggregate share.

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SECTION 1

Introduction

Lecture 4 · source PDF pp. 1–12

Most consumer products differ in buyers' eyes. Physical features, branding, service and reputation can all matter. Four questions guide the choice of a demand model:

1. **Perception:** which product characteristics enter utility?
2. **Agreement:** do consumers rank the products in the same order?
3. **Quantity:** how do individual demand curves differ; is unit demand appropriate?
4. **Variety:** does one person choose a single product or enjoy consuming several?

The examples make different assumptions natural:

Purchase	Brand choice	Quantity/variety assumption
Car	One car among alternatives	Heterogeneous tastes; unit demand is useful
Beer or soft drinks	A discrete brand can be selected	The chosen brand's quantity can vary
Wine	Several varieties may be bought	Variable quantities and love of variety

Thus *discrete choice* and *unit demand* are separate assumptions.

The course develops four tasks: distinguish modelling approaches; analyse horizontal positioning; analyse quality choice and entry; and introduce empirical demand models*. The central trade-off is between **serving more tastes** and **relaxing price competition**. The empirical section is identified as skipped in the overview but is retained.

SECTION 2

Views on product differentiation

Lecture 4 · source PDF pp. 13–37

2.1 Two modeling approaches

Lecture 4 · source PDF pp. 14–22

The *characteristics approach* defines preferences over attributes rather than merely over a fixed list of named products. It provides a way to evaluate a new product or a change in design: the product is located in a previously specified attribute space. Examples include location, sweetness, spiciness, speed, storage or service quality.

In a common application, consumers have heterogeneous ideal points and make a *discrete choice* among products and possibly an outside option. Unit demand, when imposed, adds the restriction that each consumer buys at most one unit. Discrete brand choice can also coexist with variable quantity.

The *representative-consumer approach* instead uses one consumer, or identical consumers, who buys variable amounts of several products. The linear-quadratic demand system in Lecture 2 and the CES love-of-variety model in Lecture 3 are examples. A representative consumer can also have preferences over underlying characteristics: goods are bundles of attributes, and consumption reflects the utility and implicit prices of those attributes. The characteristics approach is therefore not logically restricted to discrete choice.

	Heterogeneous discrete choice	Representative consumer
Individual purchase	One selected option; quantity may be fixed or variable	Quantities of several products
Source of aggregate demand	Distribution of tastes or choice shocks	Substitution in one utility function
Lecture examples	Hotelling and quality ladders	Quadratic utility and CES

In a characteristics-based representative-consumer model, each good is a bundle of attributes. In an interior allocation, the marginal utility of an attribute, measured in money units, equals its *implicit attribute price*. This is a different use of characteristics from choosing a single point in a discrete product set. The slides note its relevance to empirical demand analysis but do not pursue it further in the course.

The modelling choice should fit the purchase occasion. Smooth aggregate demand can arise from a population of discrete buyers even though no individual buys a fraction of a car.

2.2 Horizontal/vertical differentiation

Lecture 4 · source PDF pp. 23–37

Two rankings, with two different purposes

1. **Preference definition, at equal prices.** Disagreement over the preferred product is horizontal differentiation; unanimous preference for one product is vertical differentiation. These correspond to variety and quality.
2. **Market-structure definition, at marginal-cost prices.** Products are vertically differentiated in the technology-adjusted sense when every consumer prefers the same product with each priced at its own marginal cost. This includes the cost side and is the relevant ranking for the natural-oligopoly argument.

A higher-quality product may cost much more to produce. Agreement about which good is better at the *same* price does not guarantee agreement about which is better at *different marginal-cost prices*. The second definition matters when asking whether every potential entrant can find a profitable niche.

Real goods often combine the two dimensions. Everyone may prefer more speed and more battery life, yet consumers can value those improvements differently. If one design is faster and another has a longer battery life, there need not be a unanimous overall ranking.

Case 1: coffee differentiation

1. **Why a commodity can differentiate.** Origin, bean type and taste can distinguish coffee; specialty roasters such as Starbucks made these attributes salient. Competitions, protected trademarks and supply-chain disputes reinforce differentiation. The slides mention that analogous differentiation in tea is older.
2. **Historical example.** Fazenda Esperança won Brazil's Cup of Excellence for the 2006 harvest. A 60 kg bag sold for close to US$2,000 in a January 2007 online auction, more than ten times the commodity price; the winning bidders were from Japan and Taiwan, China.
3. **Lesson.** Some growers move up a perceived quality ladder and also differentiate tastes, while the commodity segment remains important. Real goods can mix vertical and horizontal attributes.

Review question 1

Question. In which industries is product differentiation important? Give two examples.
Answer. Smartphones differ in operating systems, cameras, battery life and design; bank deposits differ in service, branch convenience, trust and contractual features. CPUs are another lecture example. Explain the relevant characteristics and whether buyers agree on their ranking, rather than treating any brand difference as automatically vertical.

SECTION 3

Horizontal product differentiation

Lecture 4 · source PDF pp. 38–126

Product specifications are now strategic choices. We first hold prices fixed, isolating the incentive to attract a larger market share. We then let location choices affect a subsequent pricing game. Changing that continuation game can reverse the location incentives or make a pure equilibrium fail to exist.

3.1 A simple location model

Lecture 4 · source PDF pp. 39–59, 272–273

Model 1: Fixed-price location choice and minimum differentiation

Environment and timing. A unit mass of consumers is uniformly distributed on $[0,1]$. Consumer x has ideal point x. Two firms simultaneously choose locations $l_i \in [0,1]$, then consumers choose a seller. Price is regulated at $\overline{p} > c$, with constant marginal cost c. Utility from buying at i is $r - \overline{p} - \tau|x - l_i|$, and r is high enough for full coverage. Identical-location ties are split equally.

Distance may literally mean travel, or it may be a mismatch in spiciness, wine sweetness or a newspaper's political position. The mismatch parameter $\tau > 0$ measures how costly it is to substitute away from one's ideal variety.

Lecture derivation: Demand and the minimum-differentiation equilibrium

For $l_1 < l_2$, the indifferent consumer lies between the firms and solves

$$r - \overline{p} - \tau(\widehat{x} - l_1) = r - \overline{p} - \tau(l_2 - \widehat{x}),$$

so $\widehat{x} = (l_1 + l_2)/2$. Therefore

$$Q_1 = \frac{l_1 + l_2}{2}, \qquad Q_2 = 1 - \frac{l_1 + l_2}{2}.$$

The positive common margin makes maximizing profit equivalent to maximizing market share. For an arbitrary rival location l_j,

$$\pi_i(l_i, l_j) = (\overline{p} - c) \begin{cases} (l_i + l_j)/2, & l_i < l_j, \\ 1/2, & l_i = l_j, \\ 1 - (l_i + l_j)/2, & l_i > l_j. \end{cases}$$

At $l_1 = l_2 = 1/2$, either firm obtains half the market. A unilateral move left obtains $(l_i + 1/2)/2 < 1/2$; a move right obtains $1 - (l_i + 1/2)/2 < 1/2$. Neither deviation is profitable.

Lecture derivation: Demand and the minimum-differentiation equilibrium (continued)

No distinct-location pair is an equilibrium: a firm can move towards its rival without crossing it and increase its market share. Nor can both firms coincide away from 1/2: a small move towards the larger side wins more than half the market. Hence the unique location pair is

$$l_1^* = l_2^* = \frac{1}{2}. \tag{3.1}$$

Lesson 1: minimum differentiation with fixed prices

1. **Equilibrium:** two firms co-locate at $l_1 = l_2 = 1/2$ in the covered, fixed-price model.
2. **Mechanism:** the common margin is fixed, so each firm maximizes market share. Moving toward consumers has no adverse price-competition effect.
3. **Limits:** the result is a two-firm statement. Common increasing mismatch costs and suitable regular taste distributions support the median intuition, but three co-located firms are not in equilibrium and endogenous prices change incentives.

3.1.1 The social location problem

The planner values a better match, whereas firms at a fixed common margin value market share. Since everybody buys one unit, total valuation and production cost are fixed. For two products, minimize aggregate travel/mismatch cost.

Lecture derivation: Why the efficient locations are $1/4$ and $3/4$

For $0 \le l_1 \le l_2 \le 1$ and $\widehat{x} = (l_1 + l_2)/2$,

$$TC(l_1, l_2) = \tau \Bigg[\int_0^{l_1} (l_1 - x)\,\mathrm{d}x + \int_{l_1}^{\widehat{x}} (x - l_1)\,\mathrm{d}x + \int_{\widehat{x}}^{l_2} (l_2 - x)\,\mathrm{d}x + \int_{l_2}^{1} (x - l_2)\,\mathrm{d}x \Bigg].$$

Evaluating the four areas gives

$$TC = \frac{\tau}{4}\left(3l_1^2 + 3l_2^2 - 2l_1 l_2 - 4l_2 + 2\right).$$

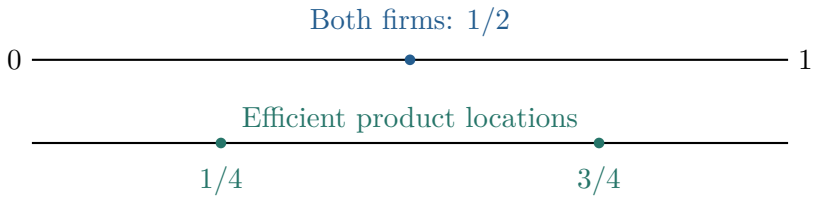
The FOCs are

$$6l_1 - 2l_2 = 0, \qquad 6l_2 - 2l_1 - 4 = 0,$$

so

$$l_1^{\text{eff}} = \frac{1}{4}, \qquad l_2^{\text{eff}} = \frac{3}{4}.$$

The Hessian of the expression in parentheses is $\left(\begin{smallmatrix} 6 & -2 \\ -2 & 6 \end{smallmatrix}\right)$, with eigenvalues 4 and 8. It is positive definite, establishing the global minimum on the ordered feasible region. Total transport cost falls from $\tau/4$ with co-location at the centre to $\tau/8$ at the efficient two locations.



Firm ownership, outlet count and location are distinct

If two firms merely co-locate, replacing them by one central outlet leaves travel unchanged and can save a duplicated firm-level fixed cost. If one owner operates two outlets, efficient spacing is feasible, but full coverage at a fixed price makes total operating revenue independent of their locations. Profit maximization alone does not necessarily select the socially efficient pair. Whether a merger saves costs also depends on whether fixed costs attach to firms or outlets. These conditions are needed to interpret the lecture's merger discussion.

Minimum differentiation is not robust to every change in the environment. Three co-located firms, for example, can each gain by moving slightly to one side and serving nearly half the market rather than one third. More importantly, location may influence a later price game, which is the next model.

3.2 The linear Hotelling model

Lecture 4 · source PDF pp. 60–105, 274–281

Model 2: Linear Hotelling location-then-price competition

Environment and timing. Consumers remain uniform on $[0,1]$ with unit mass and utility $r - p_i - \tau|x - l_i|$. First firms simultaneously choose locations; then they observe both locations and simultaneously choose prices. Marginal cost is c. Full coverage is maintained in the relevant pricing comparisons. We seek pure-strategy subgame-perfect outcomes and must therefore check the price subgames after location choices.

3.2.1 Which consumers are indifferent?

Fix distinct locations $l_1 < l_2$, set $L = l_2 - l_1$ and $S = l_1 + l_2$. If the marginal buyer lies between firms, indifference gives

$$\begin{aligned} r - p_1 - \tau(\widehat{x} - l_1) &= r - p_2 - \tau(l_2 - \widehat{x}), \\ 2\tau\widehat{x} &= \tau S + p_2 - p_1, \end{aligned}$$

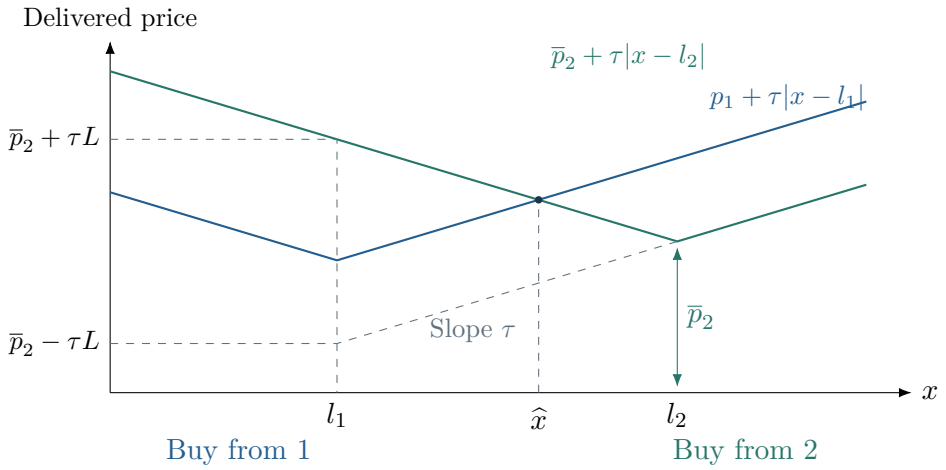
so

$$\widehat{x} = \frac{S}{2} + \frac{p_2 - p_1}{2\tau}. \tag{3.2}$$

The requirement $l_1 \le \widehat{x} \le l_2$ is equivalent to

$$p_2 - \tau L \le p_1 \le p_2 + \tau L. \tag{3.3}$$

It is not sufficient just to require $0 \le \widehat{x} \le 1$: the formula used the expressions for distance *between* the two firms.



Course figure: consumer choice in linear Hotelling. Source PDF p. 75. A consumer chooses the lower delivered price. The two V-shaped curves cross at $\widehat{x} = S/2 + (\overline{p}_2 - p_1)/(2\tau)$. The dashed extension illustrates the slope calculation; its continuation outside the relevant segment is not an additional delivered-price schedule. The two price thresholds make $\widehat{x}$ reach l_1 or l_2.

3.2.2 Why linear transport creates jumps in demand

At $p_1 = p_2 + \tau L$, the consumer at l_1 is indifferent. Every consumer to its left is also indifferent. Indeed, for $x < l_1$,

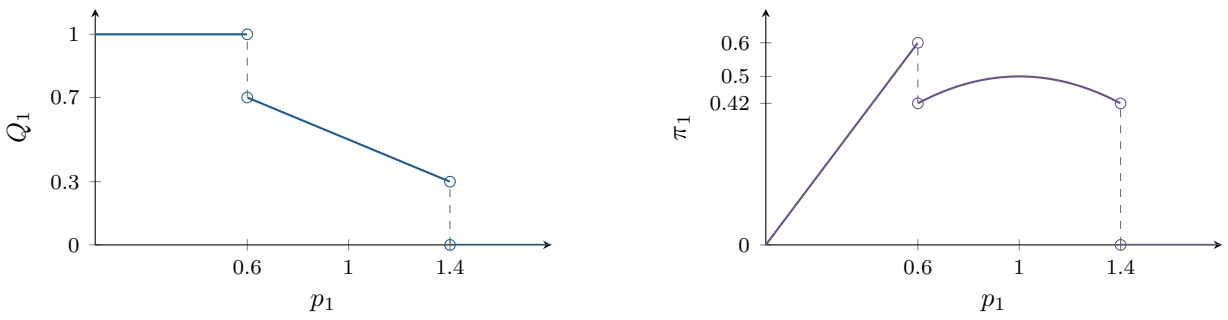
$$p_1 + \tau(l_1 - x) = p_2 + \tau L + \tau(l_1 - x) = p_2 + \tau(l_2 - x).$$

The delivered-price difference is constant for this whole group. A tiny increase in p_1 sends them all to firm 2. Likewise, a sufficiently large price cut can win all buyers at once.

Away from the threshold equalities, firm 1's demand is

$$Q_1 = \begin{cases} 0, & p_1 > p_2 + \tau L, \\ S/2 + (p_2 - p_1)/(2\tau), & |p_1 - p_2| < \tau L, \\ 1, & p_1 < p_2 - \tau L. \end{cases} \tag{3.4}$$

At equality a positive mass can be indifferent; a tie rule specifies its allocation. The strict-inequality limits are what matter for a deviation that undercuts by an arbitrarily small positive amount. Unlike the linear case, quadratic travel costs make delivered-price differences vary continuously with x throughout the consumer interval, eliminating these masses of indifferent consumers at separated locations.



Course figure: demand and profit jumps. Source PDF pp. 77–85. The profit panel redraws the two competing peaks. The demand panel makes the source of each jump explicit. Hollow points denote one-sided limits, since the allocation of indifferent consumers requires a tie rule.

For illustration, $l_1 = 0.3$, $l_2 = 0.7$, $\tau = 1$, $c = 0$ and $p_2 = 1$. Hollow points show the one-sided limits at tie thresholds; a tie rule determines the value at the threshold itself. The sharing branch peaks at $p_1 = 1$, but a price just below 0.6 gives a higher profit by winning the whole market.

3.2.3 Local price optimization and the competing all-market deviation

Profit is $(p_1 - c)Q_1$. On the sharing branch, it is a concave quadratic. On the all-market branch, it is $p_1 - c$. The whole function can have two competing peaks separated by a jump, so the sharing-branch FOC is not a global optimization certificate.

Lecture derivation: Compare the sharing optimum with the undercutting supremum

On the sharing branch,

$$\frac{\partial \pi_1}{\partial p_1} = \frac{S}{2} + \frac{p_2 - p_1}{2\tau} - \frac{p_1 - c}{2\tau} = 0,$$

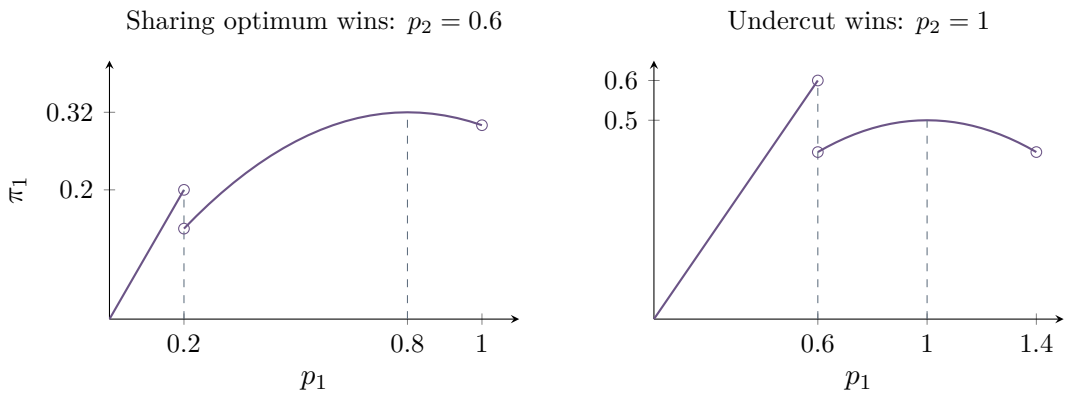
which gives the local candidate

$$p_1^D(p_2) = \frac{p_2 + c + \tau S}{2}, \qquad \pi_1^D = \frac{[p_2 - c + \tau S]^2}{8\tau}.$$

This is admissible only if its price lies on the sharing branch. To win the entire market, firm 1 can choose $p_1 = p_2 - \tau L - \varepsilon$, $\varepsilon > 0$. The limiting profit as $\varepsilon \downarrow 0$ is

$$\pi_1^U = p_2 - c - \tau L.$$

Thus a possible best-response switch is governed by comparing π_1^D with π_1^U, not by imposing the FOC everywhere. In the relevant feasible sharing range, a higher rival price can make a large undercut attractive, causing the best response to jump down. Exact attainment at the boundary depends on the tie rule. The globally defined price game therefore need not have strategic complementarity even though the local sharing response has positive slope.



Reading the course's two-peak profit diagram. Source PDF pp. 77–85; the two panels explain its global-deviation comparison with $l_1 = 0.3, l_2 = 0.7, \tau = 1, c = 0$. On the left, the sharing peak is 0.32 versus an undercutting limit of 0.2. On the right, the undercutting limit 0.6 beats the sharing peak 0.5. Thus a higher rival price can switch the preferred branch. Hollow points show limits rather than a specified tie allocation.

3.2.4 When do the two local candidates constitute an equilibrium?

Lecture derivation: Solve prices and check both firms' global deviations

The sharing FOCs imply

$$p_1 = \frac{p_2 + c + \tau S}{2}, \qquad p_2 = \frac{p_1 + c + \tau(2 - S)}{2}.$$

Solving yields

$$p_1^* = c + \frac{\tau}{3}(2 + S), \qquad p_2^* = c + \frac{\tau}{3}(4 - S). \tag{3.5}$$

The corresponding demand and profit candidates are

$$Q_1^* = \frac{2 + S}{6}, \quad Q_2^* = \frac{4 - S}{6}, \qquad \pi_1^* = \frac{\tau}{18}(2 + S)^2, \quad \pi_2^* = \frac{\tau}{18}(4 - S)^2.$$

Firm 1's all-market undercutting supremum is

$$\begin{aligned} p_2^* - c - \tau L &= \frac{\tau}{3}(4 - l_1 - l_2) - \tau(l_2 - l_1) \\ &= \frac{2\tau}{3}(2 + l_1 - 2l_2). \end{aligned}$$

Requiring its candidate profit to be at least this amount gives

$$(2 + l_1 + l_2)^2 \geq 12(2 + l_1 - 2l_2). \tag{3.6}$$

Likewise firm 2 can undercut p_1^* by τL, obtaining limiting profit

$$p_1^* - c - \tau L = \frac{2\tau}{3}(1 + 2l_1 - l_2).$$

Its no-undercutting condition is therefore

$$(4 - l_1 - l_2)^2 \geq 12(1 + 2l_1 - l_2). \tag{3.7}$$

Why the endpoints are special. At $l_1 = 0, l_2 = 1$, the jumps have sizes $l_1 = 0$ and $1 - l_2 = 0$, so demand is continuous and the usual covered endpoint pricing problem is well behaved. At interior locations the jump sizes are positive. Solving local FOCs alone can therefore miss a profitable all-market deviation. In the maintained covered model, when the separated-location pure-price equilibrium exists it is the unique sharing solution below.

For distinct locations these are the lecture's price-equilibrium existence conditions in the maintained covered market. Where the pure equilibrium exists, it is the sharing solution above. For coincident locations, products are identical and the price equilibrium is $p_1 = p_2 = c$.

As a check, impose symmetric locations $l_1 = 1 - l_2$. Then $S = 1$, both prices are $c + \tau$, and both inequalities reduce to

$$9 \geq 36(1 - l_2) \quad \Longleftrightarrow \quad l_2 \geq \frac{3}{4}.$$

Equivalently $l_1 \leq 1/4$ and the distance is *at least* $1/2$. Closer distinct symmetric products have no pure price equilibrium.

A profitable deviation missed by the FOCs

Let $c = 0$, $\tau = 1$, $l_1 = 0.3$ and $l_2 = 0.7$. The sharing FOCs give $p_1 = p_2 = 1$ and each profit $1/2$. But firm 1 can charge just below $1 - 0.4 = 0.6$, capture everybody, and obtain profit close to $0.6 > 0.5$. The candidate fails a feasible global deviation even though both local derivatives are zero.

3.2.5 Location incentives and the pure-equilibrium difficulty

Where the sharing equilibrium exists, the reduced location profits are those displayed above. A unilateral change holds the other firm's location fixed:

$$\frac{\partial \pi_1^*}{\partial l_1} = \frac{\tau}{9}(2 + S) > 0, \qquad \frac{\partial \pi_2^*}{\partial l_2} = -\frac{\tau}{9}(4 - S) < 0.$$

The left firm moves right and the right firm moves left, into the region where pure price equilibria fail. Although profit is constant if *both* firms move symmetrically and preserve $S = 1$, that coordinated change is not the unilateral deviation used to test equilibrium.

A pure-strategy subgame-perfect equilibrium must prescribe a pure price equilibrium after every location pair. Some distinct pairs have no such equilibrium, so the unrestricted two-stage game has no pure-strategy subgame-perfect equilibrium. This does not rule out mixed pricing, alternative equilibrium concepts or richer relocation dynamics. It also does not mean that *every* fixed-location price game lacks an equilibrium: the endpoints and sufficiently separated pairs are well behaved.

Lesson 2: endogenous differentiation need not yield a pure prediction

1. **Local pricing is insufficient:** a firm can prefer a discrete undercut to the sharing-branch FOC solution.
2. **Location incentives can undermine the continuation:** where a pure price equilibrium exists, firms tend to move together and can enter a region with no pure price equilibrium.
3. **Scope:** the unrestricted linear-transport location-then-price game lacks a pure-strategy subgame-perfect prediction. This does not exclude mixed strategies or imply failure at every fixed location pair.

3.2.6 Exercise 1: Hong Kong's Mid-Levels escalators

Lecture 4 · source PDF pp. 93–104

The exercise imagines a one-kilometre slope from Des Voeux Road at 0 to Conduit Road at 1, with 100,000 uniformly distributed inhabitants. Normalize their mass to one, so model quantities and profits can be multiplied by 100,000 for actual daily totals. Won-Ton (firm 1) is at 0 and Too-Chow (firm 2) is at 1. Each person buys at most one bowl of soup, $c = 0$, and r is high enough for full coverage.

Before the escalators, travel to firm 1 costs tx and travel to firm 2 costs $(t + \tau)(1 - x)$, $t, \tau > 0$. After their installation, the hypothetical costs are tx and $t(1 - x) + f$, $f > 0$. The fee is a modelling device in this exercise, not a claim about actual escalator charges. The historical installation referred to is October 1993.

The questions ask for indifferent buyers, prices, profits, the effects of τ and f, the condition for both firms to remain active, and a comparison of the two regimes.

Questions to answer.

1. **Before installation:** find the indifferent consumer, prices and profits; explain why Too-Chow can gain when climbing becomes more costly.
2. **After installation:** repeat the calculation, find the positive-demand/positive-price condition, and explain the effect of a lower access fee.
3. **Compare regimes:** show that Too-Chow loses even with free access; test Won-Ton's gain at $t = 2, f = 3$ for $\tau = 2$ and $\tau = 4$.

Lecture derivation: Before installation: asymmetric travel slopes

Indifference is

$$r - tx^o - p_1 = r - (t+\tau)(1-x^o) - p_2,$$

so

$$x^o = \frac{t+\tau-p_1+p_2}{2t+\tau}, \qquad 1-x^o = \frac{t+p_1-p_2}{2t+\tau}.$$

The price FOCs from $p_1 x^o$ and $p_2(1-x^o)$ yield

$$p_1 = \frac{t+\tau+p_2}{2}, \qquad p_2 = \frac{t+p_1}{2}.$$

Solve the two linear equations:

$$p_1^o = t + \frac{2\tau}{3}, \qquad p_2^o = t + \frac{\tau}{3}.$$

At the price optimum, each demand equals its price divided by $2t+\tau$. Therefore

$$\pi_1^o = \frac{(3t+2\tau)^2}{9(2t+\tau)}, \qquad \pi_2^o = \frac{(3t+\tau)^2}{9(2t+\tau)}. \tag{3.8}$$

For Too-Chow,

$$\frac{\partial \pi_2^o}{\partial \tau} = \frac{2(3t+\tau)(2t+\tau) - (3t+\tau)^2}{9(2t+\tau)^2} = \frac{(3t+\tau)(t+\tau)}{9(2t+\tau)^2} > 0.$$

A larger uphill penalty changes the slope of relative travel costs: demand is less price-sensitive, softening price competition. Under full coverage this competition effect can outweigh the shop's less favourable access.

Lecture derivation: After installation: a fixed access penalty

The indifferent buyer solves

$$r - tx^s - p_1 = r - t(1-x^s) - f - p_2, \quad \Longrightarrow \quad x^s = \frac{t+f-p_1+p_2}{2t}.$$

Price FOCs now give

$$p_1 = \frac{t+f+p_2}{2}, \qquad p_2 = \frac{t-f+p_1}{2},$$

so

$$p_1^s = t + \frac{f}{3}, \qquad p_2^s = t - \frac{f}{3},$$

$$\pi_1^s = \frac{(3t+f)^2}{18t}, \qquad \pi_2^s = \frac{(3t-f)^2}{18t}. \tag{3.9}$$

Lecture derivation: After installation: a fixed access penalty (continued)

Both firms have positive price and demand if $f < 3t$. Within that region,

$$\frac{\partial \pi_2^s}{\partial f} = \frac{f - 3t}{9t} < 0.$$

Unlike τ, the fixed penalty f does not change the slope of demand with respect to the price difference. It makes firm 2 uniformly less attractive, so a reduction helps it. For $f \geq 3t$, solve the corner pricing game rather than extrapolating a negative price or quantity.

Lecture derivation: Compare regimes and explain the opposite effects

Even free access after installation reduces Too-Chow's profit relative to the pre-installation regime:

$$\pi_2^o - \pi_2^s|_{f=0} = \frac{\tau(3t + 2\tau)}{18(2t + \tau)} > 0.$$

The old asymmetric travel slopes had softened competition; the new common slope intensifies it. A positive fee further reduces Too-Chow's attractiveness.
For Won-Ton, set $t = 2$, $f = 3$ as requested. Then

$$\pi_1^o - \pi_1^s = \frac{4(\tau + 3)^2}{9(\tau + 4)} - \frac{9}{4} = \frac{16\tau^2 + 15\tau - 180}{36(\tau + 4)}.$$

At $\tau = 2$ this difference is $-43/108$, so installation benefits Won-Ton. At $\tau = 4$ it is $17/36$, so installation hurts it. The fixed-fee regime gives firm 1 a relative access advantage, but replacing a large slope difference also intensifies competition. Which effect dominates depends on parameters.

3.3 The quadratic Hotelling model

Lecture 4 · source PDF pp. 106–126, 282

Model 3: Quadratic Hotelling and the differentiation trade-off

Environment and timing. Keep the location-then-price timing and uniform consumers, but replace travel cost with $\tau(x - l_i)^2$. Marginal cost is c and full coverage is maintained. First restrict product locations to $[0, 1]$.

3.3.1 Indifference and the price equilibrium

Lecture derivation: Quadratic costs give a continuous cutoff

For $l_1 < l_2$, indifference gives

$$p_1 + \tau(\widehat{x} - l_1)^2 = p_2 + \tau(\widehat{x} - l_2)^2.$$

Expand the squares and cancel $\widehat{x}^2$:

$$2\tau\widehat{x}(l_2 - l_1) = \tau(l_2^2 - l_1^2) + p_2 - p_1.$$

Thus, with $L = l_2 - l_1$, $S = l_1 + l_2$,

$$\widehat{x} = \frac{S}{2} + \frac{p_2 - p_1}{2\tau L}. \tag{3.10}$$

Lecture derivation: Quadratic costs give a continuous cutoff (continued)

The delivered-price difference is affine and strictly monotone in x over the whole line, so quantities are this cutoff clipped to $[0, 1]$. Demand changes continuously at the clipping points.

For a sharing equilibrium, the price FOCs are

$$\widehat{x} - \frac{p_1 - c}{2\tau L} = 0, \qquad 1 - \widehat{x} - \frac{p_2 - c}{2\tau L} = 0.$$

Solving gives

$$p_1^* = c + \frac{\tau}{3} L(2 + S), \qquad p_2^* = c + \frac{\tau}{3} L(4 - S), \qquad \widehat{x}^* = \frac{2 + S}{6}. \tag{3.11}$$

The continuous demand with its flat zero/all-market branches yields the regular price equilibrium for separated feasible locations. For coincident products, prices are c; the displayed separated-location formula tends to that limit as $L \downarrow 0$.

For given locations, the equilibrium profits are

$$\widehat{\pi}_1 = \frac{\tau}{18} L(2 + S)^2, \qquad \widehat{\pi}_2 = \frac{\tau}{18} L(4 - S)^2. \tag{3.12}$$

Increasing the gap L relaxes price competition, while the sum S shifts the demand division. These are reduced profits: firms optimize locations anticipating the price response, rather than differentiating profits while holding prices fixed.

3.3.2 Location choice and the two competing effects

Lecture derivation: Restricted and unrestricted product locations

Holding the other firm's location fixed, differentiate (3.12):

$$\frac{\partial \widehat{\pi}_1}{\partial l_1} = \frac{\tau}{18}(2+S)(l_2 - 3l_1 - 2),$$
$$\frac{\partial \widehat{\pi}_2}{\partial l_2} = \frac{\tau}{18}(4-S)(4 + l_1 - 3l_2).$$

For $0 \le l_1 < l_2 \le 1$, the first is negative and the second positive. Each firm moves outwards, giving the endpoint equilibrium

$$l_1^* = 0, \qquad l_2^* = 1, \qquad p_i^* = c + \tau, \qquad \pi_i^* = \frac{\tau}{2}.$$

This is maximal differentiation *within the permitted interval.*
If the feasible product range is extended, an interior location candidate instead solves

$$l_2 - 3l_1 - 2 = 0, \qquad 4 + l_1 - 3l_2 = 0.$$

Solving, without imposing symmetry before differentiation, gives

$$l_1^* = -\frac{1}{4}, \qquad l_2^* = \frac{5}{4}.$$

At this pair each own-location second derivative is $-\tau/2$, confirming the local maximum condition. The covered price continuation gives $p_i = c + 3\tau/2$ and $\pi_i = 3\tau/4$. The familiar unrestricted-location result uses the maintained fully covered formulation; a finite reservation value and arbitrarily remote deviations require the appropriate participation and global-continuation checks.

Market-size effect versus competition effect

Moving towards consumers' ideal points can attract more demand: the *market-size effect* pulls products together. Moving away from the rival makes consumers less ready to switch in response to a small price difference: the *competition effect* pushes products apart. With quadratic costs and the stated uniform tastes, the second force pushes firms to the endpoints of $[0, 1]$. With a larger feasible range, finite outward positions balance the forces; products need not run to infinity.

Lesson 3: equilibrium differentiation reflects a balance

1. **Market-size effect:** moving toward consumers improves the match and can expand a firm's sales; it draws products together.
2. **Competition effect:** moving away from a rival reduces substitution and relaxes price competition; it pushes products apart.
3. **Balance and model dependence:** quadratic transport and locations in $[0, 1]$ give the endpoints. Extending the feasible range gives finite outward positions $-1/4, 5/4$ in the covered benchmark; taste distributions, mismatch costs and residual market power can change this balance.

3.3.3 More products and the circle

With more than two firms, an interval creates boundary asymmetries and complicates location analysis. Joining the endpoints gives a circle, where equidistant locations are a natural symmetric candidate. Lecture 3 used that configuration to study entry. Suitable quadratic-cost circle models can support equidistant locations in a location-then-price equilibrium; a symmetric price equilibrium also exists at equidistant positions with linear costs in the maintained Salop setting. The circle is especially useful for entry comparisons, while "minimal" versus "maximal" differentiation is less naturally defined there.

Review question 2

Question. What brings firms together in product space, and what makes them differentiate?
Answer. The market-size effect encourages serving a large part of the taste distribution by moving towards buyers. The competition effect encourages differentiation to soften the price response of competing firms. To predict a location, specify the continuation pricing game and compare both effects.

SECTION 4

Vertical product differentiation

Lecture 4 · source PDF pp. 127–187

The horizontal model distributes ideal product locations across consumers. Here all consumers agree that higher quality is better, but differ in willingness to pay for it. Product choice is still strategic: firms first select quality, then compete in prices. The section then studies why a narrow enough range of tastes can limit the number of viable firms even with tiny entry costs.

4.1 Quality choice

Lecture 4 · source PDF pp. 128–164

4.1.1 Consumer preferences, income and the outside option

Model 4: Quality choice followed by price competition

Environment and timing. Firm i chooses quality $s_i \in [\underline{s}, \overline{s}]$, with $0 < \underline{s} < \overline{s}$. Consumers have quality tastes $\theta \in [\underline{\theta}, \overline{\theta}]$, $\underline{\theta} > 0$, distributed with density one. Their mass is $M = \overline{\theta} - \underline{\theta}$. Each buys at most one product. Baseline conditional indirect utility is $v_i = r - p_i + \theta s_i$ and the outside utility is normalized to zero. Marginal production cost is zero and does not depend on quality.

Because $\frac{\partial^2 v_i}{\partial \theta \, \partial s_i} = 1 > 0$, consumers with higher θ value a quality improvement more. They agree about quality rankings at equal prices, but a premium for quality can make low-θ buyers choose the cheaper, lower-quality good.

Lecture derivation: The slides' direct-to-indirect utility normalization

The slides write conditional direct utility for buying product i as

$$u_i(q_0; \theta) = q_0 + \theta(s_i - \overline{s})$$

and for not buying as $u_0(q_0; \theta) = q_0 - r - \theta\overline{s}$. They parametrize income as $y = r + \theta\overline{s}$. The numeraire has unit price.
If the consumer buys i, the budget gives $q_0 = y - p_i$; otherwise $q_0 = y$. Substitute into the relevant conditional utility:

$$v_i = y - p_i + \theta(s_i - \overline{s}) = r - p_i + \theta s_i,$$
$$v_0 = y - r - \theta\overline{s} = 0.$$

Thus the discrete choice is $\max\{0, \max_i(r - p_i + \theta s_i)\}$. Remaining income is accounted for once through q_0, not added twice. Feasible prices and an interior numeraire are implicit in this normalization.

In the quality-choice model take r sufficiently large that the relevant equilibrium is fully covered. The slides use the sufficient condition $p_i < r$; the actual purchase condition is that the best

available $r - p_i + \theta s_i$ is nonnegative. The two-active-firm calculation below additionally requires an interior cutoff, which is a distinct restriction.

4.1.2 Demand for two fixed qualities

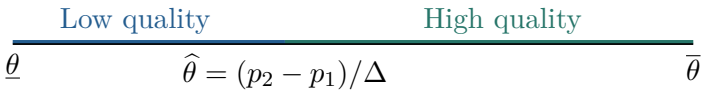
Label products so that $s_1 < s_2$ and write $\Delta = s_2 - s_1 > 0$. The indifferent consumer satisfies

$$r - p_1 + \widehat{\theta} s_1 = r - p_2 + \widehat{\theta} s_2 \quad \Longrightarrow \quad \widehat{\theta} = \frac{p_2 - p_1}{\Delta}.$$

Types below the cutoff buy low quality; those above it buy high quality. In the covered market,

$$X_1 = \begin{cases} 0, & p_2 - p_1 < \underline{\theta}\Delta, \\ (p_2 - p_1)/\Delta - \underline{\theta}, & \underline{\theta}\Delta \le p_2 - p_1 \le \overline{\theta}\Delta, \\ \overline{\theta} - \underline{\theta}, & p_2 - p_1 > \overline{\theta}\Delta, \end{cases} \qquad X_2 = M - X_1.$$

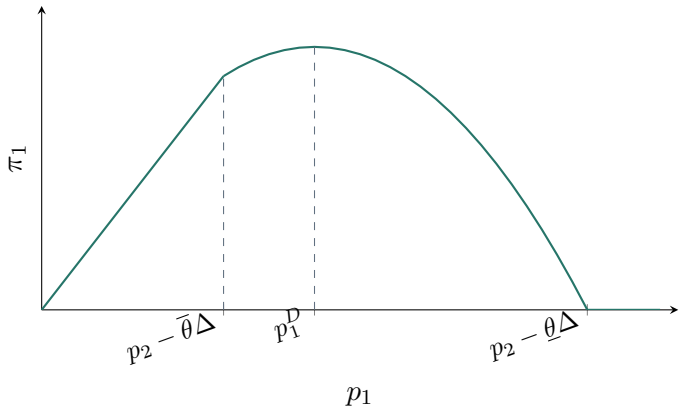
The demand is continuous at the two cutoff boundaries, unlike the separated-interior linear-Hotelling demand jumps. Positive low-quality sales require a sufficiently low price relative to high quality.



This distribution has density one: X_1 and X_2 are masses, not shares normalized to one. Dividing by M would give market shares without changing the equilibrium price FOCs.

The corresponding profit is $\pi_1 = p_1 X_1$. Its three regions have different slopes but meet continuously:

$$\pi_1 = \begin{cases} p_1 M, & p_1 < p_2 - \overline{\theta}\Delta, \\ p_1[(p_2 - p_1)/\Delta - \underline{\theta}], & p_2 - \overline{\theta}\Delta \le p_1 \le p_2 - \underline{\theta}\Delta, \\ 0, & p_1 > p_2 - \underline{\theta}\Delta. \end{cases}$$



Course figure: profit under vertical differentiation. Source PDF p. 145. The line is the all-market branch; the curve is the sharing branch; the zero branch begins once even the lowest taste prefers high quality. There is a kink but no demand/profit jump at the first boundary. The illustration uses $\underline{\theta} = 1, \overline{\theta} = 3, \Delta = 1, p_2 = 4$, with sharing optimum $p_1^D = 1.5$.

4.1.3 Stage 2: price competition for given qualities

Lecture derivation: Price FOCs, interior conditions and reduced profits

On the two-active-firm branch,

$$\pi_1 = p_1\left(\frac{p_2 - p_1}{\Delta} - \underline{\theta}\right), \qquad \pi_2 = p_2\left(\overline{\theta} - \frac{p_2 - p_1}{\Delta}\right).$$

Differentiate with qualities fixed:

$$\frac{\partial \pi_1}{\partial p_1} = \frac{p_2 - 2p_1}{\Delta} - \underline{\theta} = 0,$$
$$\frac{\partial \pi_2}{\partial p_2} = \overline{\theta} + \frac{p_1 - 2p_2}{\Delta} = 0.$$

Both own-price second derivatives are $-2/\Delta$. Rearranging the FOCs gives

$$2p_1 = p_2 - \underline{\theta}\Delta, \qquad 2p_2 = p_1 + \overline{\theta}\Delta.$$

Solve the system:

$$p_1^* = \frac{\overline{\theta} - 2\underline{\theta}}{3}\Delta, \qquad p_2^* = \frac{2\overline{\theta} - \underline{\theta}}{3}\Delta. \tag{4.1}$$

The cutoff and demands are

$$\widehat{\theta}^* = \frac{\overline{\theta} + \underline{\theta}}{3}, \qquad X_1^* = \frac{\overline{\theta} - 2\underline{\theta}}{3}, \qquad X_2^* = \frac{2\overline{\theta} - \underline{\theta}}{3}.$$

The low product has positive price and demand precisely when

$$\overline{\theta} > 2\underline{\theta}. \tag{4.2}$$

With the maintained coverage condition, the continuous piecewise demands permit the corresponding global price comparisons. Inserting prices and quantities gives the reduced quality-stage profits

$$\widetilde{\pi}_1 = \frac{(\overline{\theta} - 2\underline{\theta})^2}{9}\Delta, \qquad \widetilde{\pi}_2 = \frac{(2\overline{\theta} - \underline{\theta})^2}{9}\Delta. \tag{4.3}$$

Do not extend the interior expressions to $\overline{\theta} \leq 2\underline{\theta}$: that would assign negative low-quality demand or price. The corner is considered in the next subsection of the course.

Both prices increase with the quality gap. In particular, improving the *low* quality while keeping it below the high quality shrinks Δ, lowering *both* equilibrium prices and profits. At these prices, the cutoff does not depend on Δ, so the low firm does not increase its equilibrium customer mass. More similar products primarily intensify competition. Improving high quality instead raises the gap and benefits both firms through softer competition.

4.1.4 Stage 1: quality differentiation and quality leadership

In the interior covered model, both reduced profits are increasing in Δ. The lower-quality firm therefore chooses $\underline{s}$ and the higher-quality firm $\overline{s}$. Simultaneous quality choice has the two label permutations

$$(s_1, s_2) = (\underline{s}, \overline{s}) \quad \text{or} \quad (\overline{s}, \underline{s}).$$

The labels can be exchanged because the firms began with identical technologies; they do not mean that the low product has a higher payoff. If quality is chosen sequentially before simultaneous pricing, the first mover can choose $\overline{s}$ and secure the more profitable high-quality role; the follower chooses $\underline{s}$.

Lesson 4: differentiation can be valuable even when quality is free to improve

1. **Price stage:** both firms's reduced profits rise with the quality gap $\Delta = s_2 - s_1$ in the maintained fully covered interior model.
2. **Quality stage:** simultaneous choice yields one lowest-quality and one highest-quality firm, with either label assignment. With sequential quality choice, the first mover takes the more profitable high-quality role.
3. **Economic lesson:** differentiation can soften price competition enough that the low-quality firm prefers to remain low quality even when improvement has no production cost. This is a private incentive, not a claim that lower quality improves welfare.

The result does not establish that reducing product quality is socially desirable. A planner values the consumers' quality gain, whereas the low-quality firm's objective incorporates how a design change makes its rival compete.

Case 2: very light jets and the "battle of bathrooms"

1. **Setting.** The lecture's 2007 point-to-point air-taxi case concerns very light jets: short-haul aircraft, typically three to seven passengers plus one crew member, lighter and costing roughly half as much as conventional business jets.
2. **Product choices.** Magnum Jet chose the Adam A700; DayJet chose the Eclipse 500. The A700 cost about \$2.2 million versus \$1.5 million, offered around 7 versus 4.5 cubic metres of cabin space, and included a lavatory while the Eclipse did not.
3. **Different assessments of willingness to pay.** Magnum/Adam stressed that travellers would value an onboard bathroom. DayJet/Eclipse cited research suggesting that it mattered little on 60–90 minute trips compared with avoiding a six-hour drive, connecting flights or two days of travel for a short meeting.
4. **Lesson and limit.** The firms chose different quality/premium positions to soften direct comparison and serve different valuations. The case poses a market-share question; the lecture does not establish which strategy ultimately won, or that every traveller ranks all attributes identically.

4.2 Natural oligopolies

Lecture 4 · source PDF pp. 165–187, 283–284

Conventional entry models limit firms through fixed costs and scale economies. This subsection asks whether demand can also impose a bound that survives when entry cost becomes arbitrarily small. Keep taste dispersion distinct from market size: duplicating a population with the same θ distribution increases demand without widening $[\underline{\theta}, \overline{\theta}]$.

4.2.1 The duopoly corner and natural monopoly

Model 5: Vertical differentiation with entry and natural monopoly

For the entry illustration use $v_i = \theta s_i - p_i$, outside utility zero, zero marginal cost and $\Delta > 0$. When tastes are relatively similar, $\overline{\theta} \leq 2\underline{\theta}$, the positive-sales interior solution fails. Solve the boundary instead.

Lecture derivation: Why low quality cannot earn a positive operating profit

Consider the covered boundary prices

$$p_1 = 0, \qquad p_2 = \underline{\theta}\Delta.$$

The cutoff is $\widehat{\theta} = \underline{\theta}$, so high quality serves the whole positive-mass market and low quality has zero demand. Low quality cannot gain by raising its price; a negative price would make losses.

Given $p_1 = 0$, high quality captures all buyers for $p_2 \leq \underline{\theta}\Delta$, earning increasing revenue $p_2 M$. Above this threshold, its demand is $\overline{\theta} - p_2/\Delta$ while that quantity is positive. The right-hand revenue derivative at the threshold is

$$\overline{\theta} - \frac{2p_2}{\Delta} = \overline{\theta} - 2\underline{\theta} \leq 0.$$

Thus the high firm's boundary price is optimal; it does not want to move into the interior segment. The low firm earns zero and cannot cover any strictly positive entry cost.

It is essential to derive this corner price. Inserting $\overline{\theta} \leq 2\underline{\theta}$ into the two-active-firm prices would use an infeasible equilibrium branch.

If firms enter sequentially and can choose quality at no quality cost in this model, the first entrant chooses $\overline{s}$. A later entrant cannot offer a better quality and cannot cover a positive setup payment with an equal or inferior product. The remaining firm's price is its monopoly optimum subject to buyers' outside option. For $\overline{\theta} \leq 2\underline{\theta}$ and $v = \theta\overline{s} - p$, this is $p^m = \underline{\theta}\overline{s}$, which serves the market; the high firm's price against a hypothetical low rival was instead $\underline{\theta}\Delta$.

Lesson 5: natural monopoly with arbitrarily small positive entry cost

1. **Similar tastes can eliminate a niche:** if $\overline{\theta} \leq 2\underline{\theta}$, the low-quality rival earns zero operating profit in the covered boundary pricing game.
2. **Any positive entry cost matters:** zero operating profit cannot cover even an arbitrarily small positive setup cost. Under the sequential-entry and free-quality-choice assumptions, one highest-quality firm can be the only profitable entrant.
3. **Distinguish mechanisms:** this demand-based natural-monopoly result does not require Model 8 of Lecture 3's escalating quality expenditure. It depends on the taste range, quality set, production costs and outside-option regime.

The conclusion depends on quality costs, the quality set, the taste distribution and the outside-option/coverage regime. It is not a claim that every market with products called "high" and "low" quality must be a monopoly. With integer entry, surviving firms may have strictly positive net profits: an additional entrant can still fail to cover its cost.

Why the horizontal comparison differs. In the maintained Salop model, every finite collection of equally spaced products leaves a profitable operating niche; reducing setup cost therefore permits more entry. If incumbents relocate after entry, symmetry gives the familiar zero-profit calculation. If incumbent locations are fixed, entry can still increase as cost falls, but the analysis is more involved and active firms need not all earn zero net profit. The absence of a vertical niche is the key contrast, rather than an assumption that all differentiated markets behave alike.

4.2.2 An n-firm quality ladder and the finiteness property

Model 6: The n-firm quality ladder and the finiteness property

Suppose all n firms sell positive quantities at positive prices, and order qualities as $s_1 < \cdots < s_n$. With a covered market, define

$$\widehat{\theta}_0 = \underline{\theta}, \quad \widehat{\theta}_i = \frac{p_{i+1} - p_i}{s_{i+1} - s_i} \ (i = 1, \ldots, n-1), \quad \widehat{\theta}_n = \overline{\theta}.$$

Only products on the consumers' upper utility envelope can be active. For an active ordered ladder, firm i serves $X_i = \widehat{\theta}_i - \widehat{\theta}_{i-1}$ and earns $p_i X_i$.

Lecture derivation: Successive taste cutoffs must spread out

Differentiate each profit with the neighbouring prices and all qualities fixed. The FOCs imply

$$\begin{aligned} X_1 &= \frac{p_1}{s_2 - s_1}, \\ X_i &= \frac{p_i}{s_{i+1} - s_i} + \frac{p_i}{s_i - s_{i-1}} \quad (i = 2, \ldots, n-1), \\ X_n &= \frac{p_n}{s_n - s_{n-1}}. \end{aligned}$$

Positive active prices are ordered as $0 < p_1 < \cdots < p_n$. For an interior firm,

$$\begin{aligned} X_i &= \frac{p_i}{s_{i+1} - s_i} + \frac{p_{i-1}}{s_i - s_{i-1}} + \frac{p_i - p_{i-1}}{s_i - s_{i-1}} \\ &> \frac{p_i - p_{i-1}}{s_i - s_{i-1}} = \widehat{\theta}_{i-1}. \end{aligned}$$

Since $X_i = \widehat{\theta}_i - \widehat{\theta}_{i-1}$, this gives $\widehat{\theta}_i > 2\widehat{\theta}_{i-1}$. For the highest-quality firm,

$$X_n = \frac{p_n - p_{n-1}}{s_n - s_{n-1}} + \frac{p_{n-1}}{s_n - s_{n-1}} > \widehat{\theta}_{n-1},$$

so $\overline{\theta} > 2\widehat{\theta}_{n-1}$. At the bottom, $X_1 > 0$ implies $\widehat{\theta}_1 > \underline{\theta}$. Chaining these inequalities produces the necessary condition

$$\boxed{\overline{\theta} > 2^{n-1}\underline{\theta}.} \tag{4.4}$$

For a fixed finite ratio $\overline{\theta}/\underline{\theta}$, the count is bounded:

$$n < 1 + \log_2\left(\frac{\overline{\theta}}{\underline{\theta}}\right).$$

This is a necessary condition for an all-active positive-price ladder, not a sufficient formula for the exact free-entry count. Multiplying consumer density changes both sides of each price FOC proportionally and does not remove the cutoff restriction.

Why market size cannot create infinitely many vertical niches here

A low-quality product must be cheap enough to attract buyers despite unanimous quality rankings. For each additional product to earn a positive price and serve a segment, the taste cutoffs must spread out. A bounded willingness-to-pay range eventually runs out of room, however many consumers share those tastes. In a horizontal model, different ideal points can instead sustain distinct niches.

4.2.3 Quality-dependent marginal costs and the second definition

If marginal cost depends on quality, firm profit is $(p_i - c(s_i))X_i$. In the lecture's regular vertical ladder environment, a sufficient demand/technology restriction for the finiteness result is that $c'(s)$ remain outside $[\underline{\theta}, \overline{\theta}]$ over the feasible quality interval. With a continuous derivative on that interval, the relevant alternatives are

$$\max_s c'(s) < \underline{\theta} \qquad \text{or} \qquad \min_s c'(s) > \overline{\theta}.$$

The following proof links this restriction to unanimous rankings *at marginal-cost prices.*

Lecture derivation: Compare qualities when each product is priced at marginal cost

Take $s_2 > s_1$. The utility difference at $p_i = c(s_i)$ is

$$v_2 - v_1 = \theta(s_2 - s_1) - [c(s_2) - c(s_1)].$$

By the mean value theorem, $c(s_2) - c(s_1) = c'(\zeta)(s_2 - s_1)$ for some $\zeta \in (s_1, s_2)$.
If $\max_s c'(s) < \underline{\theta}$, then

$$v_2 - v_1 > [\theta - \underline{\theta}](s_2 - s_1) \geq 0.$$

All consumers strictly prefer the higher-quality product at its marginal-cost price.
If $\min_s c'(s) > \overline{\theta}$, then

$$v_2 - v_1 < [\theta - \overline{\theta}](s_2 - s_1) \leq 0.$$

All strictly prefer the lower-quality product at its marginal-cost price: the extra quality costs more than any type values it. Both cases yield a unanimous cost-adjusted ranking, with opposite preferred directions.

This returns to Section 2: equal-price quality rankings use preferences alone, while the market-structure definition also incorporates technology. Intermediate cost slopes can create profitable niches for different taste types and require a different entry analysis.

SECTION 5

Empirical analysis of product differentiation*

Lecture 4 · source PDF pp. 188–270

Instructor-starred section; identified as skipped in the lecture overview. The material is included in the course's sequence. Theoretical demand models restrict how buyers respond to prices and characteristics. Structural empirical analysis uses these restrictions to estimate preferences, recover markups or costs, and simulate changes in a market. Product specifications are taken as given in this section; the focus is consumer demand and pricing.

The essential chain is

$$\text{individual utility} \longrightarrow \text{choice probabilities} \longrightarrow \text{market shares} \longrightarrow \text{estimable equations.}$$

The distribution of unobserved tastes determines substitution patterns, so it is an economic assumption, not merely a convenient source of regression error.

5.1 Probabilistic choice and the logit model

Lecture 4 · source PDF pp. 189–223

5.1.1 Why individual choice can look probabilistic

Two observations motivate probabilistic choice: a person may choose differently on repeated occasions under apparently similar circumstances, and observed choices may appear intransitive. The lecture distinguishes two interpretations:

1. **Intrinsic variation:** consumers are statistically identical ex ante, but different realizations of sensations, attention or evaluation errors produce different choices ex post.
2. **Unobserved heterogeneity:** each consumer behaves deterministically, but the observer does not see all tastes, information or circumstances. Choice then appears random to the observer.

Random preferences and random decision rules are alternative modelling descriptions. A binary random-threshold rule below can reproduce the same choices as an additive random-utility model; this need not mean irrational behaviour.

Write utility for product i as

$$v_i = \overline{v}_i + \epsilon_i,$$

where $\overline{v}_i$ is systematic utility measured by the model and ϵ_i is an unobserved component, often normalized to have mean zero. In a population, these are consumer-specific realizations; write ϵ_{ij} when the consumer index j is needed. A single product shock common to all buyers is different from idiosyncratic taste heterogeneity.

Case 3: the iPhone's 2007 introduction

The 2007 iPhone launch produced queues as well as less enthusiastic reactions, even after observable consumer differences were considered. The case distinguishes two explanations.

1. **Variation within the decision-maker.** A stimulus generates a sensation or psychological state; the person compares the resulting sensations across products. Attention can fluctuate, and unfamiliar features can be evaluated with error. A new product makes this interpretation particularly relevant.
2. **Missing information for the observer.** Consumers can act deterministically while the researcher does not see their tastes or state. The omitted component can reflect unobserved attributes, unobserved state-dependent utility, measurement error, or functional misspecification.

The observable choice probability alone does not distinguish these mechanisms. Probabilistic modelling need not imply irrational consumers.

5.1.2 Binary choice: convert a utility inequality to a probability

A buyer chooses product 1 over product 2 when

$$\overline{v}_1 + \epsilon_1 > \overline{v}_2 + \epsilon_2 \quad \Longleftrightarrow \quad \epsilon_2 - \epsilon_1 < \overline{v}_1 - \overline{v}_2.$$

Let $\epsilon = \epsilon_2 - \epsilon_1$ have CDF F. With no mass of ties, the choice probability is

$$Q_1 = F(\overline{v}_1 - \overline{v}_2), \qquad Q_2 = 1 - Q_1. \tag{5.1}$$

For unit consumer mass, probabilities equal aggregate demands. For mass M, multiply by M. The same binary rule can be described as a random threshold for choosing between deterministic measured utilities.

If the *difference* ϵ is uniform on $[-L, L]$, $L > 0$, its density is $1/(2L)$ there, and

$$Q_1 = \begin{cases} 0, & \overline{v}_1 - \overline{v}_2 \le -L, \\ \dfrac{1}{2} + \dfrac{\overline{v}_1 - \overline{v}_2}{2L}, & -L < \overline{v}_1 - \overline{v}_2 < L, \\ 1, & \overline{v}_1 - \overline{v}_2 \ge L. \end{cases}$$

Uniform differences give a linear interior demand, linking probabilistic choice to the Hotelling cutoff calculation. With asymmetric locations or characteristics, the relevant support need not be symmetric. Specifying the *difference* as uniform does not say that two independent uniform product shocks have a uniform difference; their difference would instead have a different distribution.

For a logistic difference with scale $\mu > 0$,

$$F(e) = \frac{1}{1 + \exp(-e/\mu)},$$

the resulting binary logit is

$$Q_1 = \frac{1}{1 + \exp[-(\overline{v}_1 - \overline{v}_2)/\mu]} = \frac{\exp(\overline{v}_1/\mu)}{\exp(\overline{v}_1/\mu) + \exp(\overline{v}_2/\mu)}. \tag{5.2}$$

Larger μ means more dispersed taste differences and less deterministic choice. As $\mu \downarrow 0$, the higher systematic utility wins except at a tie. A linear demand model can be convenient, but the suitable empirical specification depends on the actual substitution behaviour; a logit form is not automatically correct for every market either.

5.1.3 Multinomial logit: the joint distribution matters

With more than two alternatives, pairwise difference distributions alone do not define a coherent joint utility model. The standard logit specification takes the ϵ_i i.i.d. type-I extreme value, with scale μ. A centred CDF and density are

$$F(e) = \exp\left\{-\exp\left[-\left(\frac{e}{\mu} + \gamma_E\right)\right]\right\},$$
$$f(e) = \frac{1}{\mu}\exp\left[-\left(\frac{e}{\mu} + \gamma_E\right)\right] F(e),$$

where $\gamma_E \simeq 0.5772$ is Euler's constant. The mean is zero and variance is $\pi^2\mu^2/6$. The common centring does not affect utility differences or choice probabilities.

Lecture derivation: From independent extreme-value shocks to logit shares

Conditional on $\epsilon_i = t$, product i wins when $\epsilon_j \leq t + \overline{v}_i - \overline{v}_j$ for all $j \neq i$. Independence implies

$$Q_i = \int_{-\infty}^{\infty} f(t) \prod_{j \neq i} F(t + \overline{v}_i - \overline{v}_j)\,\mathrm{d}t.$$

Let $z = \exp[-(t/\mu + \gamma_E)]$. The density element becomes $f(t)\,\mathrm{d}t = -\exp(-z)\,\mathrm{d}z$, with the integration direction reversed. Also

$$F(t + \overline{v}_i - \overline{v}_j) = \exp\{-z\exp[(\overline{v}_j - \overline{v}_i)/\mu]\}.$$

Consequently

$$\begin{aligned} Q_i &= \int_0^{\infty} \exp\left\{-z\sum_j \exp[(\overline{v}_j - \overline{v}_i)/\mu]\right\}\mathrm{d}z \\ &= \frac{1}{\sum_j \exp[(\overline{v}_j - \overline{v}_i)/\mu]} \\ &= \frac{\exp(\overline{v}_i/\mu)}{\sum_j \exp(\overline{v}_j/\mu)}. \end{aligned}$$

This derives the multinomial formula rather than assuming that the binary formula can simply be repeated independently for every pair.

For $n \geq 3$, within the lecture's i.i.d. full-support regular random-utility setting, the extreme-value family is the characteristic distribution associated with the multinomial-logit choice formula across utility values, up to the relevant location/scale normalizations. Outside that maintained class, different latent structures can generate similar observed probabilities.

5.1.4 IIA, replicated alternatives and path independence

Keep the lecture's three justifications separate:

1. **Distributional route:** independent type-I extreme-value utility shocks yield the multinomial formula through the probability integral just derived. A logistic *difference* suffices for the binary model; the multinomial model needs a joint distribution.
2. **Replication route:** expanding every type into equally many statistically identical alternatives preserves aggregate type probabilities under logit. The individual clones' probabilities change; their sum is the relevant object.

3. **Choice-consistency route:** IIA concerns relative odds, and path independence uses compatible aggregate weights. Neither means that deleting a stochastic alternative leaves every absolute probability unchanged.

The logit formula implies

$$\frac{Q_i}{Q_j} = \exp\left(\frac{\overline{v}_i - \overline{v}_j}{\mu}\right). \tag{5.3}$$

The *odds* of i versus j do not depend on the attributes or presence of other alternatives. This is independence of irrelevant alternatives (IIA). Removing another alternative generally *does* change each remaining absolute probability by changing the denominator; it preserves their odds. An alternative with a low systematic utility can still be selected with positive probability when taste shocks have full support.

The lecture also discusses expanding a choice set by making k statistically identical replicas of *every* alternative. In logit, each type's numerator and the common denominator both acquire the factor k, so its total probability aggregated over its k replicas is unchanged. Each individual replica's probability is divided by k; it is not individually unchanged. Replicating just one type instead changes the distribution of choice across types.

For a partition of alternatives into groups g, let

$$A_g = \sum_{j \in g} \exp(\overline{v}_j/\mu), \qquad I_g = \mu \log A_g.$$

If group choice uses weights $\exp(I_g/\mu) = A_g$, and conditional choice within a group uses the original logit weights, then

$$\Pr(g)\Pr(i \mid g) = \frac{A_g}{\sum_h A_h}\frac{\exp(\overline{v}_i/\mu)}{A_g} = Q_i.$$

This is the relevant path-independence interpretation: a consistent aggregate weight for each group gives the same final probabilities. An arbitrary two-stage decision rule need not reproduce logit. These odds and aggregation properties, together with standard positivity/consistency axioms, provide the lecture's axiomatic motivation for the form.

5.2 Empirical analysis of horizontal product differentiation

Lecture 4 · source PDF pp. 224–243

5.2.1 Demand inversion: from shares to a linear equation

Environment and timing. Consumers choose among n inside products and an outside option. Normalize outside systematic utility to zero and the taste-shock scale to one. Let α_i denote an observed inside share and α_0 the outside share. Product specifications are fixed.

Systematic utility is

$$\overline{v}_i = \beta x_i + \xi_i - \gamma p_i, \qquad \gamma > 0,$$

where x_i contains observed characteristics, β their coefficients, and ξ_i the utility of characteristics observed by market participants but omitted by the researcher. The price coefficient γ here is not Euler's constant γ_E.

Lecture derivation: Cancel the common logit denominator

With $D = 1 + \sum_j \exp(\overline{v}_j)$,

$$\alpha_i = \frac{\exp(\overline{v}_i)}{D}, \qquad \alpha_0 = \frac{1}{D}.$$

Taking the ratio and then logarithms gives

$$\log \alpha_i - \log \alpha_0 = \overline{v}_i = \beta x_i + \xi_i - \gamma p_i. \tag{5.4}$$

The outside share is important: it anchors the relative utility scale and the size of participation in the market. The inversion assumes positive shares; zero observed shares require an explicit sampling or modelling treatment rather than taking an undefined logarithm.

Although (5.4) is linear in the parameters, ordinary regression is generally inappropriate if high unobserved quality ξ_i raises the price firms can charge. Price then correlates with the error. Instruments must shift price while satisfying an appropriate exclusion from the unobserved demand characteristic; relevance alone is not enough. Product characteristics also require their maintained exogeneity or suitable instruments.

5.2.2 Supply: recover a markup equation from the price FOC

Assume a single-product firm, constant marginal cost $c_i = \kappa w_i + \omega_i$, and price competition. The cost shifters w_i and coefficients κ need not equal the demand characteristics and their coefficients. Profit is $(p_i - c_i)M\alpha_i$.

Lecture derivation: The single-product logit pricing equation

Holding rival prices fixed, the FOC is

$$M\alpha_i + (p_i - c_i)M\frac{\partial \alpha_i}{\partial p_i} = 0.$$

Differentiate the logit share through systematic utility:

$$\frac{\partial \alpha_i}{\partial \overline{v}_i} = \alpha_i(1 - \alpha_i), \qquad \frac{\partial \overline{v}_i}{\partial p_i} = -\gamma,$$

so $\frac{\partial \alpha_i}{\partial p_i} = -\gamma\alpha_i(1 - \alpha_i)$. Cancel $M\alpha_i > 0$ in the FOC:

$$1 - \gamma(p_i - c_i)(1 - \alpha_i) = 0.$$

The pricing equation is therefore

$$p_i = \kappa w_i + \frac{1}{\gamma(1 - \alpha_i)} + \omega_i. \tag{5.5}$$

Demand equation (5.4) and supply equation (5.5) can be estimated jointly under suitable identification conditions. After estimating demand, the implied marginal cost is price minus the model's markup.

The FOC is a maintained supply restriction, not an independent proof of existence or uniqueness of every industry equilibrium. With multiple products under common ownership, changing one price also changes profits on sister products; the single-product markup formula must then be replaced by the appropriate system of FOCs.

5.2.3 Elasticities reveal the restrictions imposed by logit

For $i \neq j$, differentiation gives $\frac{\partial \alpha_i}{\partial p_j} = \gamma \alpha_i \alpha_j$. Thus the signed price elasticities are

$$\eta_{ij} = \frac{\partial \alpha_i}{\partial p_j} \frac{p_j}{\alpha_i} = \begin{cases} -\gamma p_i (1 - \alpha_i), & i = j, \\ \gamma p_j \alpha_j, & i \neq j. \end{cases} \tag{5.6}$$

Two restrictions to assess in an application.

1. **Own-price restriction:** common linear price disutility links percentage elasticity closely to the price level when shares are small. A particular markup pattern is then built into the functional form.
2. **Cross-price restriction:** i.i.d. product shocks imply proportional diversion across alternatives. Similarity in observable characteristics does not by itself make two products closer substitutes.

If all individual shares are small, the absolute own-price elasticity is approximately γp_i. A cheaper product therefore has a smaller percentage response to a percentage price increase. The corresponding relative markup, or Lerner index, is

$$\frac{p_i - c_i}{p_i} = \frac{1}{\gamma p_i (1 - \alpha_i)}.$$

At small shares it is larger for lower-priced products, while the *absolute* markup is approximately the same $1/\gamma$. It is important to specify which meaning of markup is being used.

Cross-price elasticities for a fixed changing price p_j are the same for every other inside product i. Diversion is proportional to existing shares rather than to closeness in physical characteristics. Introducing a new product leaves the odds of every original pair unchanged.

A simple illustration of the IIA limitation

Suppose a car and a bus have equal systematic utility, giving shares 1/2 and 1/2. Add a second bus with the same measured utility and an independent logit shock. All three shares become 1/3: the new bus draws from the car as well as the original bus. If the two buses are almost the same service, that substitution pattern can be implausible. The example illustrates the model's restriction; it is not an estimated result about actual transport demand.

These restrictions follow from common linear price disutility and i.i.d. taste shocks. They can be useful benchmarks but should be assessed against actual substitution in the market being studied.

5.3 Empirical analysis of vertical product differentiation

Lecture 4 · source PDF pp. 244–250

Suppose consumer j has utility

$$v_{ij} = \theta_j s_i - p_i + \epsilon_{ij}, \qquad s_i = \beta x_i + \xi_i,$$

with $\theta_j > 0$ a scalar taste for quality and $\mathbb{E}\theta_j = 1$ as a normalization. Consumers agree that more s is better, but value an improvement differently. The scalar θ_j multiplies both observed and unobserved product quality. The mean systematic utility is

$$\mathbb{E}[\theta_j s_i - p_i] = s_i - p_i = \beta x_i + \xi_i - p_i.$$

This mean has the earlier form with price coefficient one, but it does not by itself identify aggregate shares.

Average utility is not a shortcut to average choice

If conditional shocks are logit, the share is

$$\alpha_i = \int \frac{\exp[(\theta s_i - p_i)/\mu]}{1 + \sum_k \exp[(\theta s_k - p_k)/\mu]} \, \mathrm{d}F(\theta).$$

In general this is not logit evaluated at $\mathbb{E}\theta = 1$: the choice formula is nonlinear. The distribution of θ, not merely its mean, matters. If product-specific taste shocks remain nondegenerate, even nonadjacent quality products can have cross-price substitution.

The lecture contrasts two limiting specifications. In *pure deterministic vertical choice*, one scalar θ drives sorting on the quality ladder, and local price changes shift cutoffs between adjacent *active* products. In *standard multinomial logit*, there is an idiosyncratic taste component for every alternative and all positive-share products are substitutes.

	Standard logit	Pure vertical benchmark
Heterogeneity	Product-specific taste components	One taste for quality, θ
Local substitution	All positive-share alternatives	Adjacent products on the active quality ladder
Main restriction	Proportional diversion / IIA	One-dimensional ordered sorting

The adjacency statement is local: large price changes can remove a product from the active ladder and change neighbours. A vertical model with additional independent shocks is richer than this pure benchmark and does not inherit its adjacency restriction unchanged. Both benchmarks motivate more flexible empirical specifications.

5.4 Nested logit and other extensions

Lecture 4 · source PDF pp. 251–270

5.4.1 Use nests to allow stronger substitution among related products

There are three ways to treat market segments:

1. **One ordinary logit:** simple to estimate, but every product pair inherits IIA.
2. **Separate segment models:** allow different within-segment demand but impose no substitution across segments.
3. **Nested logit:** retain cross-segment substitution and allow extra substitution within nests; the prescribed grouping still restricts relative substitution.

A new family van is likely to compete more closely with other family vans than with every SUV or small car in proportion to their market shares. Completely separate segment models impose no cross-segment substitution. Nested logit is an intermediate approach: related alternatives share a nest, allowing stronger substitution within nests while retaining cross-nest responses.

Let g index nests, I_g its products and $\delta_i = \beta x_i + \xi_i - \gamma p_i$. For the usual parameterization, $0 \le \sigma < 1$; $\sigma = 0$ gives ordinary logit. The conditional share is

$$\alpha_{i|g} = \frac{\exp[\delta_i/(1-\sigma)]}{D_g}, \qquad D_g = \sum_{j \in I_g} \exp[\delta_j/(1-\sigma)].$$

With the outside option a separate alternative,

$$\alpha_g = \frac{D_g^{1-\sigma}}{1 + \sum_h D_h^{1-\sigma}}, \qquad \alpha_i = \alpha_{i|g}\alpha_g.$$

The nesting represents a correlated utility structure; it need not mean that every consumer consciously executes that exact sequence of decisions.

Lecture derivation: Nested-share inversion and the adjusted price equation

Write $H = 1 + \sum_h D_h^{1-\sigma}$, so $\alpha_0 = 1/H$. Then

$$\frac{\alpha_i}{\alpha_0} = \exp[\delta_i/(1-\sigma)] D_g^{-\sigma}.$$

Since $\log \alpha_{i|g} = \delta_i/(1-\sigma) - \log D_g$,

$$\log \alpha_i - \log \alpha_0 = \delta_i + \sigma \log \alpha_{i|g}.$$

Consequently the estimable demand equation is

$$\log \alpha_i - \log \alpha_0 = \beta x_i - \gamma p_i + \sigma \log \alpha_{i|g} + \xi_i. \tag{5.7}$$

Both price and the within-nest share term are generally endogenous and need appropriate instruments.
For a single-product price setter, differentiating the nested share gives

$$\frac{\partial \alpha_i}{\partial p_i} = -\frac{\gamma}{1-\sigma}\alpha_i\left[1 - \sigma\alpha_{i|g} - (1-\sigma)\alpha_i\right].$$

Using $\alpha_i + (p_i - c_i)\frac{\partial \alpha_i}{\partial p_i} = 0$ gives

$$p_i - c_i = \frac{1-\sigma}{\gamma[1 - \sigma\alpha_{i|g} - (1-\sigma)\alpha_i]}. \tag{5.8}$$

At $\sigma = 0$ this reduces to the earlier logit markup. Reusing the ordinary-logit supply equation at $\sigma > 0$ would impose the wrong own-price derivative.

For a given rival j in the same nest, the cross-price derivative includes the extra term

$$\frac{\partial \alpha_i}{\partial p_j} = \gamma\alpha_i\left[\alpha_j + \frac{\sigma}{1-\sigma}\alpha_{j|g}\right] \qquad (i \ne j,\ i, j \in I_g),$$

whereas across nests it is $\gamma\alpha_i\alpha_j$. This explains the additional within-nest substitution while making clear that levels also depend on the relevant shares. Nesting still restricts relative substitution within each prescribed group.

Case 4: nested logit in the US car market

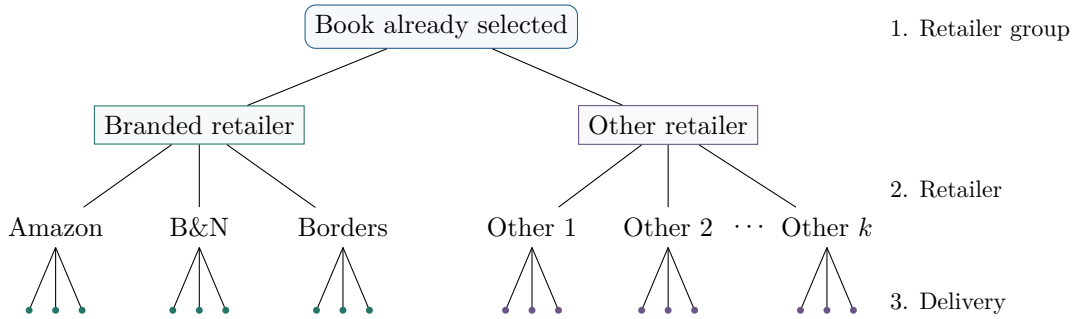
1. **Data and choice structure.** Goldberg (1995) uses Consumer Expenditure Survey microdata for the US automobile market to estimate nested logit models of car choice.
2. **Supply.** The automobile industry is modelled as a differentiated-product oligopoly; the estimated demand and supply structure connect preferences to equilibrium outcomes.
3. **Application.** Counterfactual simulations examine voluntary export restraints and exchange-rate pass-through. These policy calculations depend on the chosen structural model and its identification.

5.4.2 Retailer differentiation even for the same physical book

A consumer may have already chosen a manufacturer's product but still choose among retailers. For expensive physical purchases, visiting sellers costs time; online comparison can reduce that cost. Retailers may nevertheless differ through service, delivery and reputation, including attributes the buyer cannot inspect before purchase.

Case 5: Internet bookshops and retailer brands

1. **Data and decision.** Smith and Brynjolfsson (2001) observe clicks from more than 20,000 shopbot users over more than two months. A book is first selected on EvenBetter.com; the study then models the offer chosen for that same book. Clicks are not completed purchases.
2. **Three choice levels.** The consumer chooses branded versus other retailers, then a particular retailer, then a delivery method. Amazon, Barnes & Noble and Borders form the branded group. The diagram below retains all three levels.
3. **Result and robustness.** In ordinary logit, a branded-retailer indicator captures a strong preference premium after controlling for price and delivery time. Alternative nested-logit specifications retain the qualitative finding, addressing concern that ordinary logit's restrictive elasticities drive it.
4. **Interpretation.** Shopbot users are likely more price-sensitive than direct-site visitors. A brand premium even among these users supports the value of retailer reputation and challenges the view that online comparison eliminates brands. It does not imply a universal premium or convert clicks into purchases.



Terminal branches represent alternative delivery methods for the chosen retailer.

Course figure: the Internet-shopbot choice tree. Source PDF p. 264. The three decision levels are group, retailer and delivery. The three branded sellers are shown explicitly; the ellipsis preserves the source's open-ended group of other retailers. Delivery branches are schematic options, not claims about the number offered by a particular seller.

5.4.3 Random coefficients when predetermined nests are too restrictive

A nest structure is helpful if the market offers a defensible grouping. Without such a grouping, consumers can instead have different characteristic coefficients. For example,

$$v_{ij} = x_i\beta_j + \xi_i - \gamma p_i + \epsilon_{ij}, \qquad \beta_j \sim F_\beta.$$

Conditional on a draw β, a logit probability may still be easy to compute. The aggregate share is

$$\alpha_i = \int \frac{\exp(x_i\beta + \xi_i - \gamma p_i)}{1 + \sum_k \exp(x_k\beta + \xi_k - \gamma p_k)} \, \mathrm{d}F_\beta(\beta).$$

Different buyers place different values on product attributes, allowing substitution to follow characteristic similarity without fixing a single set of nests. The integral generally lacks the simple share inversion of ordinary logit and is approximated by simulation during estimation. The distribution of coefficients, price endogeneity, instruments and supply assumptions still require justification; flexibility does not remove the identification problem.

The empirical section's progression is therefore purposeful: simple logit provides a transparent inversion and pricing equation, vertical sorting shows another strongly restricted substitution pattern, nesting relaxes selected restrictions, and random coefficients allow richer interactions between consumers and characteristics.

Source concordance and references

The following map uses physical pages in the supplied PDF. Progressive builds are consolidated in the indicated discussion; the original appendix calculations appear in the main text beside their models.

PDF pages	Destination	Material
1–12	§1	Introduction, modelling questions and objectives
13–22	§2	Two modelling approaches
23–31	§2	Two definitions of horizontal and vertical differentiation
32–35	§2	Case 1: coffee
36–37	§2	Review question 1
38–54	§3.1	Fixed-price location model
55–59	§3.1	Efficient locations, ownership and scope
60–75	§3.2	Linear Hotelling: timing, demand regions and discontinuities
76–85	§3.2	Local versus global price optimization
86–92	§3.2	Price existence and location-stage incentives
93–104	§3.2	Exercise 1: escalators, all questions and worked solutions
105–118	§3.3	Quadratic Hotelling prices and locations
119–124	§3.3	Robustness and circle models
125–126	§3.3	Review question 2
127–140	§4.1	Quality environment, conditional utility and coverage
141–154	§4.1	Quality-ladder demand and price equilibrium
155–158	§4.1	Quality-stage equilibrium and sequential quality choices
159–164	§4.1	Case 2: very light jets
165–178	§4.2	Entry, corner prices and natural monopoly
179–187	§4.2	N-firm quality ladder and cost-adjusted rankings
188–200	§5.1	Starred random-utility section and interpretations
201–207	§5.1	Case 3: iPhone and unobserved heterogeneity
208–216	§5.1	Binary choice, uniform and logistic differences
217–223	§5.1	Multinomial logit, distribution, axioms and aggregation
224–229	§5.2	Share inversion and identification
230–236	§5.2	Supply equation and joint estimation
237–243	§5.2	Elasticities, markups and IIA
244–250	§5.3	Vertical empirical models and comparison table
251–257	§5.4	Nested logit and share equation
258	§5.4	Case 4: Goldberg automobile study
259–261	§5.4	Retailer choice
262–268	§5.4	Case 5: Internet bookshops and choice tree
269–270	§5.4	Random coefficients
271	Back matter	Source appendix divider; derivations integrated in text
272–273	§3.1	Original appendix: efficient transport costs
274–275	§3.2	Original appendix: best-response comparison
276–280	§3.2	Original appendix: price equilibrium and undercutting checks
281	§3.2	Original appendix: nonexistence discussion
282	§3.3	Original appendix: unrestricted location calculation
283–284	§4.2	Original appendix: marginal-cost ranking proof

Lecture source and cited work

Hui Shen, *Product Differentiation*, slides dated October 21, 2026. The supplied file contains 284 physical pages, including progressive builds, worked exercises and an appendix.

The empirical cases cite Goldberg (1995), on the US automobile market, and Smith and Brynjolfsson (2001), on Internet bookshop choice. Coffee and very-light-jet figures are retained as the lecture's historical illustrations. The escalator exercise is hypothetical, including its fee assumption.

The star on empirical product differentiation follows the instructor's section heading and outline. The section is identified as skipped in the overview but included in these notes. All five cases, both review questions, the full escalator exercise and all original appendix calculations are included in their corresponding main-text sections.